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Manuscript No. 17-TIE-2621

**“Adaptive Parameter Estimation and Control Design for Robot Manipulators
with Finite-Time Convergence”**

Thank you very much for your letter dated on 11-Oct-2017, informing us of the editorial decision and comments on our paper No.17-TIE-2621 (“Title: Adaptive Parameter Estimation and Control Design for Robot Manipulators with Finite-Time Convergence”). The paper has been duly revised and the presentation has been significantly improved according to the comments made by the Associate Editor and the Reviewers. We would like to submit the revised version for the possible publication in the *IEEE Transactions on Industrial Electronics*. The authors would like to express their sincere appreciation to the Associate Editor and the Reviewers for their constructive comments and the effort and the time spent helping us to improve the quality of the paper. In the following, we provide a specific response to the comments, explaining how the paper is revised.

The details of our revision as well as point-to-point responses to the reviewers are appended below. Thank you very much for the time and effort again.

Response to To Associate Editor:

Manuscript No.17-TIE-2621

**“Adaptive Parameter Estimation and Control Design for Robot Manipulators
with Finite-Time Convergence”**

The authors would like to express their sincere appreciation to the Associate Editor for his/her time and efforts to have our work reviewed and evaluated, and the recognition of the contribution of the paper.

Comment 1: The authors should carefully consider the reviewers’ comments/suggestions and appropriately prepare the revised manuscript with exact replies to reviewers.

Response: We sincerely thank the Associate Editor for the constructive comment. Accordingly, we have carefully addressed all comments from the reviewers, and have carefully improved the presentation and quality of the revised manuscript in addition to the modifications highlighted in blue color. Specifically, more comparative studies and robustness tests have been performed to show the advantages of the proposed estimation algorithm as suggested by Reviewer 1 and Reviewer 3. More theoretical analysis and more detailed derivations have been added to improve the correctness as suggested by Reviewer 2. More comprehensive literature review and discussion have been given to highlight the novelty of the paper as suggested by Reviewer 4. Besides, experiments have also been carried out and performed to shown the validity of the proposed algorithm based on a real robot. We have also read, checked and polished the revised manuscript thoroughly and invited a native English-speaker to help us polishing this paper.

We would like to thank again the Associate Editor and anonymous reviewers for the time and effort spent in handling our manuscript, as well as many constructive comments provided for improving the presentation and quality of this manuscript. The details of our revision as well as point-to-point responses to the reviewers are appended below.

Response to Referee No. 1:

Manuscript No.17-TIE-2621

**“Adaptive Parameter Estimation and Control Design for Robot Manipulators
with Finite-Time Convergence”**

The authors would like to express their sincere appreciation to the referee for his/her time and efforts to have our work reviewed and evaluated, and the recognition of the contribution of the paper.

1. *Comments: “ This paper presents an adaptive control method for robot manipulators. The paper is generally well presented and of interest. ”*

Response: Thanks for your recognition of our work.

2. *Comments: “ The only remark of this reviewer that is significant is related to the simulation results. The authors do not mention if measurement noise and other disturbances have been added to the simulation study. This reviewer considers that in the scope of having a correct evaluation, it is important to add such noise and disturbances and an improved version of this paper should be prepared to fix this issue. ”*

Response: The authors sincerely thank the reviewer for the constructive comments. We agree that it is important to consider the measurement noises in the estimation, since the sensors data may subject to sensor noise in real applications. According to your suggestion, tests with external disturbances and noises are conducted and presented to substantiate the robustness of the proposed finite-time estimation algorithm in the revised manuscript. For your convenience, the detailed test results are given as follows, please refer to Section IV, Page 7 for more details.

To further show the effectiveness of our proposed estimation algorithm, robustness tests are conducted by adding unknown external disturbances to the system model. In some practical applications, uncertainties may inevitably exist in the measurement of end-effector position, e.g., the visual tracking system may be effected by environmental noises. Without loss of generality, in the comparative studies, we conduct the tests by adding white noises with different intensity to the measured end-effector position. The external disturbance is chosen to be $d(t) = \sigma w(t)$, where σ is the degree of intensity, and $w(t)$ is the white Gaussian noise with the power of 1 dBW. For fair comparisons, the parameters in the adaptive estimation are chosen to be the same.

The simulation results are depicted as shown in Figs. 2-5. The disturbances added on the measured end-effector position are shown in Fig. 1, Fig. 3 and Fig. 5, with σ chosen to be $\sigma = 0.005$, $\sigma = 0.01$ and $\sigma = 0.02$. It can be observed from the figures that the maximum deviations added on the measured positions are about ± 2 cm, ± 4 cm and ± 6 cm, respectively. The parameters estimation performance are shown in Fig.2, Fig.4 and Fig.6. We can see from the figures that, the parameters of three groups are observed with

satisfactory convergence rate and accuracy, even in the presence of external disturbances. This has verified the robustness of the proposed estimation algorithm.

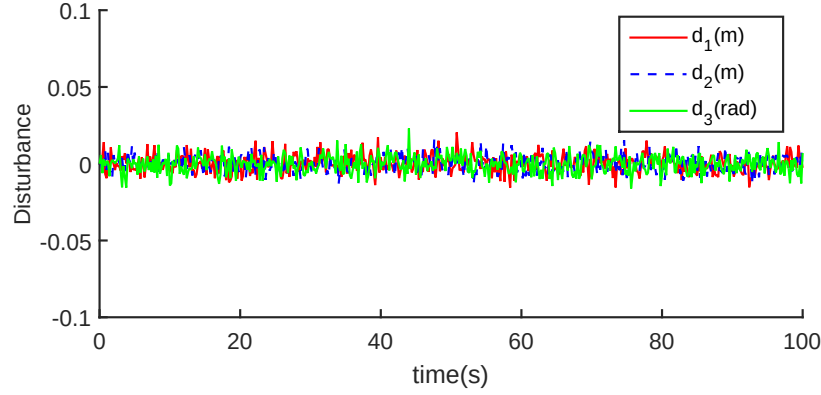


Figure 1: The added disturbances on the end-effector position $\sigma = 0.005$

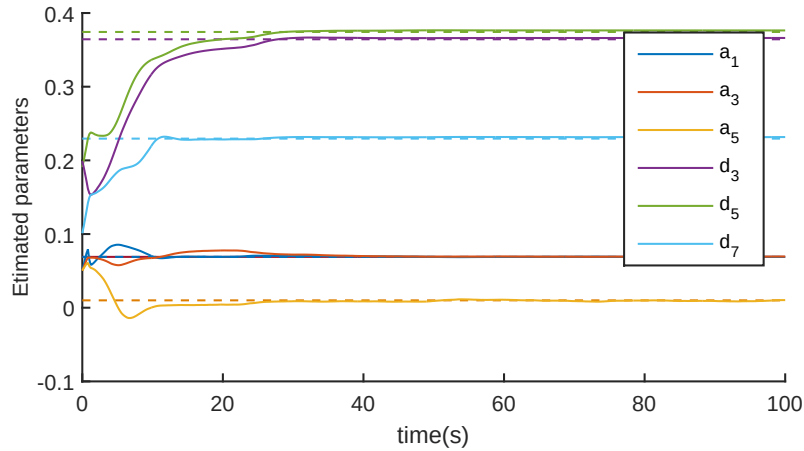


Figure 2: The kinematics estimation performance with the disturbance $\sigma = 0.005$

As suggested, the above robustness tests have been presented in Section IV, Page 7 of the revised manuscript.

3. *Comments: Minor correction: - “existed” in the second line of abstract should be changed to something else, “from” before eq. (1) should probably be “form”.*

Response: Thanks for your helpful comments to improve our paper. According to your suggestion, these typos have been all corrected. We have carefully proof-read the paper and tried our best to correct similar typos in the manuscript. The modifications are highlighted in blue color and please refer to Section I, Page 1 for more details.

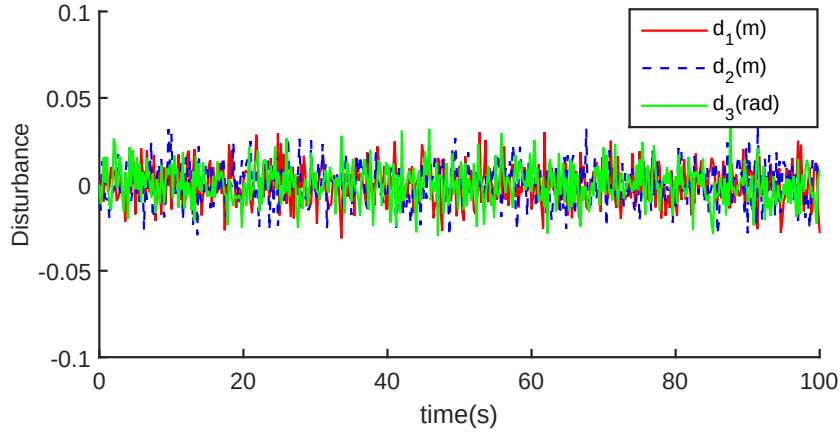


Figure 3: Disturbances added on the end-effector position with $\sigma = 0.01$

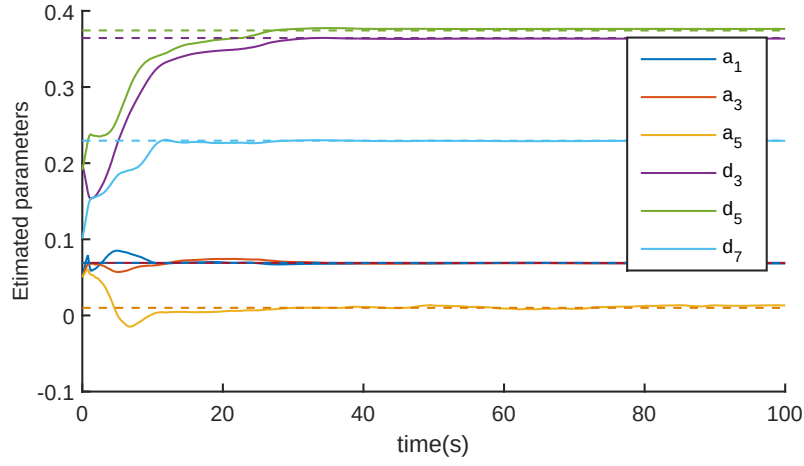


Figure 4: The kinematics estimation performance with the disturbance $\sigma = 0.01$

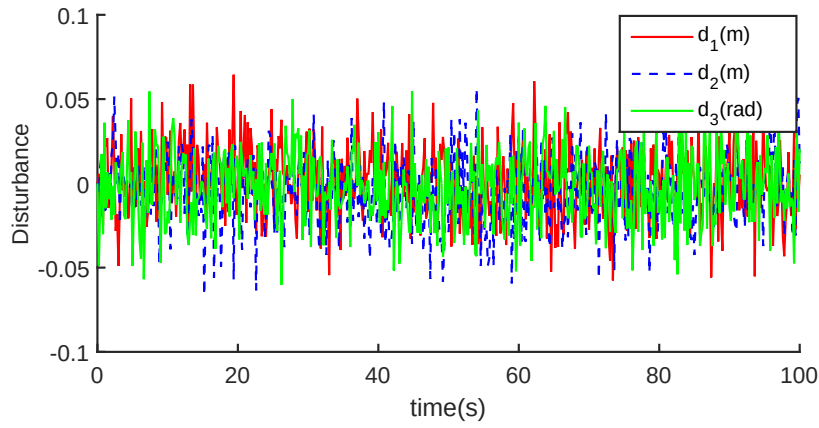


Figure 5: Disturbances added on the end-effector position with $\sigma = 0.02$

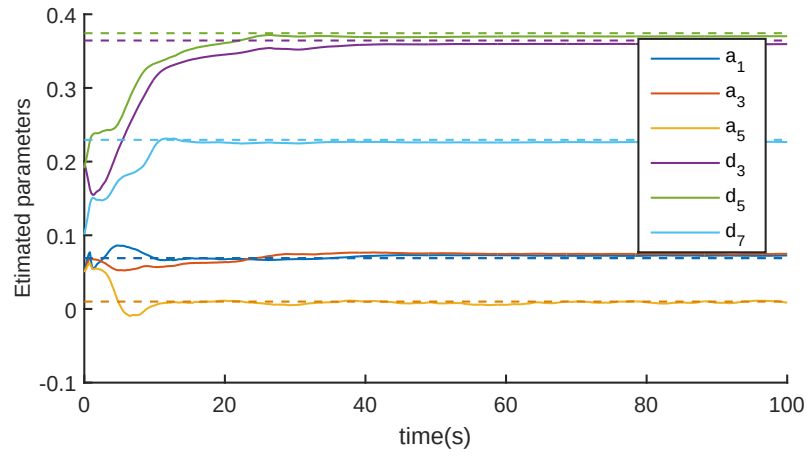


Figure 6: The kinematics estimation performance with the disturbance $\sigma = 0.02$

Response to Referee No. 2:

Manuscript No.17-TIE-2621

**“Adaptive Parameter Estimation and Control Design for Robot Manipulators
with Finite-Time Convergence”**

The authors would like to express their sincere appreciation to the referee for his/her time and efforts to have our work reviewed and evaluated, and the recognition of the contribution of the paper.

1. *Comments: “The paper aims to propose an adaptive control design for parameter estimation and tracking errors of robot manipulators within finite time. Stability analysis is performed. Simulation on a 7-DOF robot manipulator is presented. My major concerns are as follows. Page 2: The statement after (1) that ‘Taking the partial defferentiation of (1) with respect to’ is not correct. ”*

Response: Thanks for your constructive comments to improve our paper. As suggested by the reviewer, the statement of robot Jacobian matrix has been corrected and updated in the revised manuscript. For your convenience, the revised text is listed as below. Please see Section II, Page 2 of the revised manuscript for details.

The update text of the statement

Then, the relationship between the joint velocity $\dot{q}$ and end-effector velocity $\dot{x}$ can be obtained by taking the partial differentiation of $f_{kine(q)}$ as

$$\dot{x} = \frac{\partial f_{kine(q)}}{\partial q} \dot{q} = J(q, \theta) \dot{q}$$

2. *Comments: “Page 4, line 5 of Remark 2: The authors claim that ‘ D would be positive definite if ...’. This is not true. It is clear from (11) that $D(t)$ includes $\int_0^t e^{-\theta(t-r)}$, which may not enough to assure that D would be positive definite when time t so larger. This point is important to improve the estimation. Unfortunately, the positive definiteness of D may not hold true. Moreover, the proof of Theorem 1 relies on $D > \varepsilon I$ (Page 7, line 2 after (29)). As the reviewer pointed out, the positive definiteness of D may not hold true and thus $D > \varepsilon I$ does not hold. Hence, the proof of Theorem 1 is open.”*

Response: Thanks for your constructive comments, which are appreciated. In the adaptive parameter estimation, the verification of persistence excitation (PE) condition is important, since the PE condition needs to be fulfilled to guarantee the error convergence. However, the online verification of PE condition is a non-trivial issue, especially for the nonlinear robot system. In this paper, the matrix $\mathcal{D}$ is introduced for the verification of PE condition, i.e., if the regressor vector R_f in equation (11) is PE, then the matrix $\mathcal{D}$ is positive definite, with $\lambda_{\min}(\mathcal{D}) > \sigma > 0$. The proof is detailed as follows.

From the definition of PE condition, we can obtain that $\int_t^{t+T} R_f^T(\tau) R_f(\tau) d\tau \geq \varepsilon I$, which also can be expressed as $\int_{t-T}^t R_f^T(\tau) R_f(\tau) d\tau \geq \varepsilon I$, for $t > T > 0$. Thus we can obtain

that, if R_f satisfies the PE condition, the relationship $\int_{t-T}^T R_f^T(\tau)R_f(\tau)d\tau \geq \varepsilon I$ will hold for $t > T > 0$.

Then, let us consider the integration interval $\tau \in [t-T, t]$. Since $t-\tau \leq T$, we can obtain that $e^{-\vartheta(t-\tau)} \geq e^{-\vartheta T} > 0$ according to the monotonicity of the exponential function. Then the following inequality can be derived.

$$\int_{t-T}^t e^{-\vartheta(t-\tau)} R_f^T(\tau)R_f(\tau)d\tau \geq \int_{t-T}^t e^{-\vartheta T} R_f^T(\tau)R_f(\tau)d\tau \geq e^{-\vartheta T} \varepsilon I$$

In addition, we can also derive that for all $t > T > 0$,

$$\int_0^t e^{-\vartheta(t-\tau)} R_f^T(\tau)R_f(\tau)d\tau > \int_{t-T}^t e^{-\vartheta(t-\tau)} R_f^T(\tau)R_f(\tau)d\tau$$

From the above analysis, we have that

$$\mathcal{D} = \int_0^t e^{-\vartheta(t-r)} R_f^T(r)R_f(r)dr > e^{-\vartheta T} \int_{t-T}^t R_f^T(r)R_f(r)dr \geq e^{-\vartheta T} \varepsilon I$$

This implies that $\mathcal{D}$ is positive definite and its minimum eigenvalue satisfies $\mathcal{D}(t) > \sigma$ with $\sigma = e^{-\vartheta T} \varepsilon$.

The above discussions have been summarized as a remark and presented in Section III, Page 4 of the revised manuscript and the added text is listed below for your convenience.

The update text of Remark 2

Remark 2: From the definition of PE condition, we can obtain that $\int_t^{t+T} R_f^T(r)R_f(r)dr \geq \varepsilon I$, which also equals to $\int_{t-T}^t R_f^T(r)R_f(r)dr \geq \varepsilon I$, for $t > T > 0$. Thus we can derive that, if R_f satisfies the PE condition, the relationship $\int_{t-T}^t R_f^T(r)R_f(r)dr \geq \varepsilon I$ will hold for $t > T > 0$. Then, let us consider the integration interval $r \in [t-T, t]$. Since $t-r \leq T$, we can obtain that $e^{-\vartheta(t-r)} \geq e^{-\vartheta T} > 0$ according to the monotonicity of the exponential function. Then we can derived that, $\int_{t-T}^t e^{-\vartheta(t-r)} R_f^T(r)R_f(r)dr \geq \int_{t-T}^t e^{-\vartheta T} R_f^T(r)R_f(r)dr \geq e^{-\vartheta T} \varepsilon I$. In addition, $\int_0^t e^{-\vartheta(t-r)} R_f^T(r)R_f(r)dr > \int_{t-T}^t e^{-\vartheta(t-r)} R_f^T(r)R_f(r)dr$ holds for all $t > T > 0$. From the above analysis, we have $\mathcal{D} = \int_0^t e^{-\vartheta(t-r)} R_f^T(r)R_f(r)dr > e^{-\vartheta T} \int_{t-T}^t R_f^T(r)R_f(r)dr \geq e^{-\vartheta T} \varepsilon I$. This implies that $\mathcal{D}$ is positive definite and its minimum eigenvalue satisfies $\mathcal{D}(t) > \sigma$ with $\sigma = e^{-\vartheta T} \varepsilon$.

3. *Comments: "Page 5, (18). There is a typo in (18). Moreover, it is not very clear how to get (18). The statement above (18) is so confused. It seems that v_c and y_c should satisfy some specified relationship to give the representation of U_c as $U_c = v_c A_c y_c^T$."*

Response: The authors sincerely thank the reviewer for the constructive comment. According to your suggestions, the typos in equation (18) have been corrected. Additionally, the statement above (18) has been carefully rewritten and the presentation has been improved. Generally, the formula $\mathcal{U}_c = \mathcal{V}_c \mathcal{A}_c \mathcal{Y}_c^T$ can be obtained by using the singular-value-decomposition (SVD) method, which is widely employed in matrix operations. The $\mathcal{V}_c$ is an orthogonal matrix whose columns are eigenvectors of $\mathcal{U}_c \mathcal{U}_c^T$, $\mathcal{Y}_c$ is an orthogonal

matrix whose columns are eigenvectors of $\mathcal{U}_c^T \mathcal{U}_c$, and $\mathcal{A}_c$ is a diagonal matrix of the form $\mathcal{A}_c = \text{diag}\{\sigma_1, \dots, \sigma_r\}$, with $r = \text{rank}(\mathcal{U}_c)$. Note that the matrix $\mathcal{V}_c$ and $\mathcal{Y}_c$ are unitary matrices such that the relationships $\mathcal{V}_c \mathcal{V}_c^T = I$ and $\mathcal{Y}_c \mathcal{Y}_c^T = I$ hold.

As suggested by the reviewer, the above analysis for singular-value-decomposition has been added in Section III, Page 5 of the revised manuscript. The added text is listed below for your convenience.

the update text of the SVD method

Then, let us employ the SVD for the matrix $\mathcal{U}_c$ as $\mathcal{U}_c = \mathcal{V}_c \mathcal{A}_c \mathcal{Y}_c^T$, where $\mathcal{V}_c$ is an orthogonal matrix whose columns are eigenvectors of $\mathcal{U}_c \mathcal{U}_c^T$, $\mathcal{Y}_c$ is an orthogonal matrix whose columns are eigenvectors of $\mathcal{U}_c^T \mathcal{U}_c$, and $\mathcal{A}_c$ is a diagonal matrix of the form $\mathcal{A}_c = \text{diag}\{a_1, \dots, a_n\}$. Note that the matrix $\mathcal{V}_c$ and $\mathcal{Y}_c$ are unitary matrices such that the relationships $\mathcal{V}_c \mathcal{V}_c^T = I$ and $\mathcal{Y}_c \mathcal{Y}_c^T = I$ hold.

4. Comments: “Page 5, (19): The derivation of the first equality of (19) should be in detailed.”

Response: Thanks for your helpful comment to improve our paper. According to your suggestion, the derivation of equation (19) ((20) in the revised version) has been detailed formulated in the revised manuscript. For your convenience, the added text is listed as below, please refer to Section III, Page 5 for more details.

the update text of the derivation

Since $\mathcal{V}_c$ satisfies that $\mathcal{V}_c^T \mathcal{V}_c = I$, multiplying $\mathcal{V}_c^T$ on both sides of $\mathcal{U}_c = \mathcal{V}_c \mathcal{A}_c \mathcal{Y}_c^T$, we can obtain that $\mathcal{V}_c^T \mathcal{U}_c = \mathcal{A}_c \mathcal{Y}_c^T$. Then, multiplying $\mathcal{U}_c$ on both side of (19), we have

$$\begin{aligned} Q\mathcal{U}_c &= \mathcal{Y}_c (\mathcal{A}_c + e^{-\delta t} \eta I)^{-1} \mathcal{V}_c^T \mathcal{U}_c \\ &= \mathcal{Y}_c (\mathcal{A}_c + e^{-\delta t} \eta I)^{-1} \mathcal{A}_c \mathcal{Y}_c^T \\ &= \mathcal{Y}_c \text{diag}\left(\frac{a_1}{a_1 + e^{-\delta t} \eta}, \dots, \frac{a_n}{a_n + e^{-\delta t} \eta}\right) \mathcal{Y}_c^T \end{aligned}$$

Note that in above derivations, the fact $\mathcal{A}_c$ is a diagonal matrix is used.

5. Comments: “Page 5, above (22): ‘and for zero eigenvalue a_j , $\lim_{t \rightarrow \infty} \dots = 1$ ’?”

Response: Thanks for your constructive comments to improve our paper. According to your suggestion, the above mentioned equation has been updated and the typo has been corrected. In the revised version, the definition of $F(t)$ is modified to “ $F(t) = \mathcal{Y}_c \text{diag}\left(\frac{e^{-\delta t} \eta}{a_1 + e^{-\delta t} \eta}, \dots, \frac{e^{-\delta t} \eta}{a_n + e^{-\delta t} \eta}\right) \mathcal{Y}_c^T$,” and the corresponding formulation is modified to “for the nonzero eigenvalue a_i , the term $\frac{e^{-\delta t} \eta}{a_i + e^{-\delta t} \eta}$ would converge to zero when $t \rightarrow \infty$, and for zero eigenvalue a_j , $\lim_{t \rightarrow \infty} \frac{e^{-\delta t} \eta}{a_j + e^{-\delta t} \eta} = 1$ ”. For your convenience, the modified text is listed below, and more details could be found in Section III, Page 5 of the revised manuscript.

The update text

Let us define an auxiliary term $F(t)$ as

$$F(t) = \mathcal{Y}_c \text{diag}\left(\frac{e^{-\delta t} \eta}{a_1 + e^{-\delta t} \eta}, \dots, \frac{e^{-\delta t} \eta}{a_n + e^{-\delta t} \eta}\right) \mathcal{Y}_c^T$$

Then (21) can be rewritten as

$$Q(t)\mathcal{U}_c(t) = I - F(t)$$

Note that for the nonzero eigenvalue a_i , the term $\frac{e^{-\delta t \eta}}{a_i + e^{-\delta t \eta}}$ would converge to zero when $t \rightarrow \infty$, and for zero eigenvalue a_j , $\lim_{t \rightarrow \infty} \frac{e^{-\delta t \eta}}{a_j + e^{-\delta t \eta}} = 1$.

6. *Comments: "Page 8, (34): How to obtain the last two inequality of (34) is not so clear. Please give a detailed derivation."*

Response: Thanks for your constructive comments to improve our paper. According to your suggestion, more details about the derivation of equation (34) ((35) in the revised version) have been added in the revised manuscript. For your convenience, the revised text is listed below, and please refer to Page 8, left column for more details.

the update of the derivation

According to (22) and considering that $F\Phi_r = \Phi_r - QB_c$, $B_c = U_c\Phi_r$, (34) can be rewritten as

$$\begin{aligned} \dot{L}_d &= -e_s^T K_1 e_s - (\hat{\Phi}_r - \Phi_r)^T \frac{\hat{\Phi}_r - QB_c}{\|\hat{\Phi}_r - QB_c\|} - K_2 \frac{e_s^T e_s}{\|e_s\|} \\ &\leq -(\hat{\Phi}_r - F\Phi_r - QB_c)^T \frac{\hat{\Phi}_r - QB_c}{\|\hat{\Phi}_r - QB_c\|} - K_2 \frac{e_s^T e_s}{\|e_s\|} \\ &\leq -(\hat{\Phi}_r - QB_c)^T \frac{\hat{\Phi}_r - QB_c}{\|\hat{\Phi}_r - QB_c\|} + \|F\Phi_r\| - K_2 \|e_s\| \\ &\leq -\|\hat{\Phi}_r - QB_c\| + \|F\Phi_r\| - K_2 \|e_s\| \\ &\leq -\|\Phi_r - QB_c + \hat{\Phi}_r - \Phi_r\| + \|F\Phi_r\| - K_2 \|e_s\| \\ &\leq -\|F\Phi_r - \tilde{\Phi}_r\| + \|F\Phi_r\| - K_2 \|e_s\| \\ &\leq -\|\tilde{\Phi}_r\| - K_2 \|e_s\| + 2\|F\Phi_r\| \\ &\leq -\mu_d \sqrt{L_d} + \rho_d \end{aligned} \tag{1}$$

where $\mu_d = \min\{\sqrt{2/\lambda_{\max}(\Gamma_d^{-1})}, K_2 \sqrt{2/\lambda_{\max}(M)}\}$, $\rho_d = 2\|F\|\|\Phi_r\|$. Since F and Φ_r are bounded and $\lim_{t \rightarrow \infty} F = 0$, there exists a compact set $\Omega := \left\{ \tilde{\Phi}_r | L_d(\tilde{\Phi}_r) \leq (c/\mu_d)^2 \right\}$, such that $\forall \tilde{\Phi}_r \notin \Omega$, $\dot{L}_d < 0$ in term of the result in [32], with c being a positive constant satisfies that $c > \gamma_d |\rho_d|$, $0 < \gamma_d < 1$.

Response to Referee No. 3:

Manuscript No.17-TIE-2621

“Adaptive Parameter Estimation and Control Design for Robot Manipulators with Finite-Time Convergence”

The authors would like to express their sincere appreciation to the referee for his/her time and efforts to have our work reviewed and evaluated, and the recognition of the contribution of the paper.

1. *Comments: “In this paper, an adaptive scheme to estimate the kinematic and dynamic parameters with finite-time convergence was proposed and simulation results were given. An adaptive estimation technique was proposed for a robot manipulator before [20]. In the previous study [20], information of estimation errors were used. In this paper, a set of auxiliary filtered matrices were designed for the previous adaptive control law [20]. So, the main adaptive control law was derived before [20] and there is no new derivation of controller. For this reason, originality of this paper is poor and this paper is not suitable for the journal. ”*

Response: The authors sincerely thank the reviewer for the constructive comments. As the reviewer pointed out, the adaptive estimation technique is important for robot manipulators to achieve better control performance and has been widely investigated in previous literatures, e.g., in [20], an adaptive parameters estimation method was designed to estimate the unknown robot dynamics by using the information of the estimation errors. Very **different from** the previous work, in this paper, we developed an **adaptive parameter** estimation algorithm for **both** the robot dynamic parameters and **kinematic parameters**. The kinematic uncertainties widely exist in robot control, for example, when a robot is equipped with different types of end-effectors or to grasp a tool with unknown length, these kinematic parameters will alter and lead to the unknown manipulator Jacobian matrix. Thus, it is desirable to achieve fast and accurate estimation of robot kinematic parameters. However, among the literatures about adaptive robot control, few studies have concerned with the estimation of robot kinematic parameters with finite-time (FT) convergence. **For this purpose**, an adaptive FT estimation algorithm has been developed to estimate the unknown robot kinematic parameters, which has shown superior performance than conventional adaptive estimation technique in term of the estimation speed. By using the adaptive FT estimation algorithm, the estimated kinematic parameters are observed with satisfactory convergence performance in both simulation and experiment studies.

On the other hand, in this paper, the Newton-Euler (NE) dynamic model has been employed in dynamic parameters identification to improve the computational efficiency. In the literature [20], the estimation algorithm is based on a conventional Lagrange dynamics model, which requires explicit mathematical form of the dynamics and is cost expensive

in calculation. In addition, the amount of calculation of Lagrange model will increase significantly with the increase of the DOFs and this may lead to the problem called ‘curse of dimensionality’. Hence, Lagrange model has been merely used for identification of a high DOF robot, such as Baxter with 7 DOFs of each arm. In comparison to the Lagrange model, the NE model is of more computational efficiency. For instance, the computational complexity of the NE model for a N DOF robot is $O(N)$, which means that the computational load is a linear function of N and is far less than the Lagrange model. Therefore, comparing with [20], the NE model based identification developed in this paper is computationally more efficient and more suitable for real-time control applications.

Another contribution to be highlighted is the improved accuracy of the estimated parameters by using the model reduction method. It is well known that, persistence excitation (PE) condition should be satisfied to fulfill the convergence of the estimated parameters. However, the conventional NE dynamic model may not be able to guarantee the PE condition, due to the restriction of the robot mechanism and lack of sensors information. For example, in the NE regressor of Baxter robot, some columns of the matrix may be linear dependent of others, and thus the matrix $R_f^T R_f$ in the PE condition may not be full rank. As a result, the PE condition would not be satisfied and the estimated parameters may not converge to their real values. To remedy the potential singularity problem encountered in the conventional NE dynamics model, in this paper we develop a singular value decomposition based model reduction method. The main idea is to build a minimum set of inertial parameters by rearranging and identifying the inertial parameters that do not effect the robot dynamics, using the singular value decomposition (SVD) method. By using the proposed model reduction method, PE condition can be fulfilled and the estimation results show the fast and accurate convenience of estimated parameters.

Thank the reviewer again for the inspiring comment so that the authors have the opportunity to present the novelties and contributions in a clear manner. As suggested, the above discussion has been presented in Section I, Page 2 of the revised manuscript.

2. *Comments: “Comparison results between the proposed and conventional PD controller are given in Figures 9, 10 and 11. The comparison results do not show the advantage of the proposed controller because PD controller is simple. The tracking performances of the proposed, PID and the adaptive controller [20] should be depicted and significant advantages of proposed should be given. Experimental results will increase the quality of this paper. ”*

Response: The authors sincerely thank the reviewer for the constructive comments, which are appreciated. In this paper, we proposed a finite time parameters estimation scheme to identify the kinematic and dynamic parameters of the robots with enhanced convergence rate and accuracy. The conventional PID control methods have achieved great success in robot manipulators control. However, satisfactory control performance may not be obtained using the PID control because the unknown robot dynamics act as a prominent external disturbance and may lead to degradation of the control performance or even incur

instability. Model based adaptive control (MBAC) has been thus introduced to handle the effect of the uncertain robot dynamics. For most of MBAC based robot systems, however, gradient decent based adaptive laws were designed to minimize the tracking errors and parameters prediction errors. Although elegant mathematical results have been obtained with enhanced steady-state performance, the potentially poor transient performance still limit the practical applications, due to the strong coupling between the tracking error and the observer error. To improve the estimation performance, in this paper, we design a finite time robot control scheme with a leakage term driven by parameter estimation errors, such that precise and fast estimation of robotic parameters can be obtained by using the information estimation error. Comparing with traditional control methods, e.g., PID control or the gradient descent based MBAC, the proposed adaptive estimation algorithm is able to guarantee the finite-time identification with fast parameters convergence and better control performance.

On the other hand, in this paper, the NE dynamic model was used in the estimation of robot dynamic parameters. Generally, the NE dynamic model costs less calculation than Lagrange model when modeling a high DOFs robot. The NE model based estimation algorithm proposed in this paper can therefore attain more computational efficient in comparison with the Lagrange model based adaptive estimation method in [20]. It should be emphasized that, this is not a trivial issue since the PE condition of the NE model may not be guaranteed due to the restriction of the robot mechanism and lack of sensors information. For this reason, an SVD based model reduction method is proposed to relax the requirement of PE condition and the estimation results show that the estimated parameters can accurately converge to their real values. This issue has not been addressed in the previous works.

According to your suggestion, comparisons have been conducted based on the PID control and a MBAC [16] in the revised manuscript. Additionally, the dynamic parameters estimated using the proposed algorithm and adaptive method in [20] have also been detailed performed in Table III, to show the performance of the proposed estimation algorithm. Moreover, experiment results of the estimation have also been conducted and performed in Section IV, Page 7 of the revised version. For your convenience, the comparisons results are depicted as below.

the update of the comparative results

To further verify the effectiveness of proposed algorithm, comparison studies have been carried out based on a PID controller and a model based adaptive controller (MBAC) [16]. The comparison results are depicted in Figs.7-14. As seen from the figures, although the stable tracking can be obtained by using all three types of controller, the tracking errors of the PID control (dashed-dotted line ‘-.’) are relatively large, since the PID controller is seriously effected by the robot dynamics such as inertial and Coriolis torque. The MBAC controller has achieved well steady-state tracking performance (dashed line ‘—’) after a period time of adaptation, but the performance of transient state is much lower than the

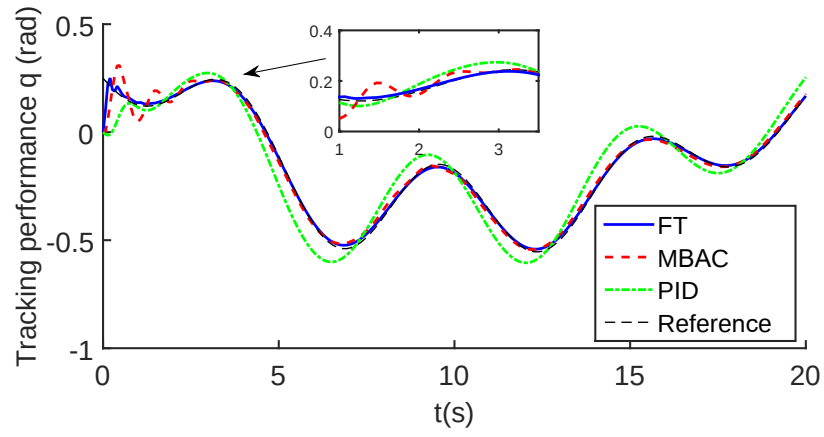


Figure 7: Tracking performance of Joint 1

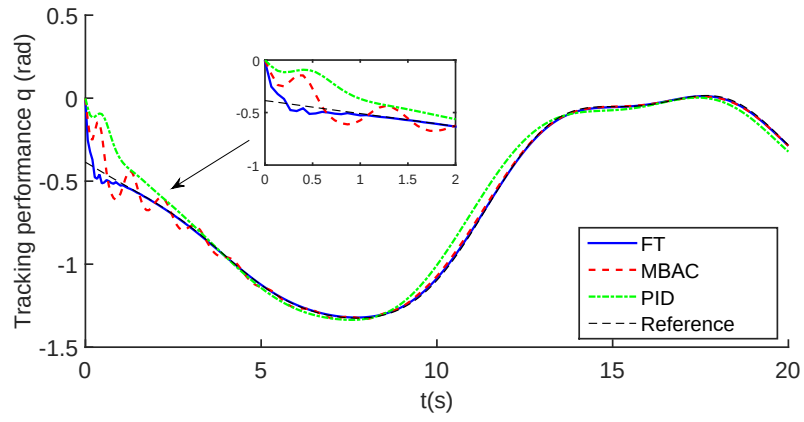


Figure 8: Tracking performance of Joint 2

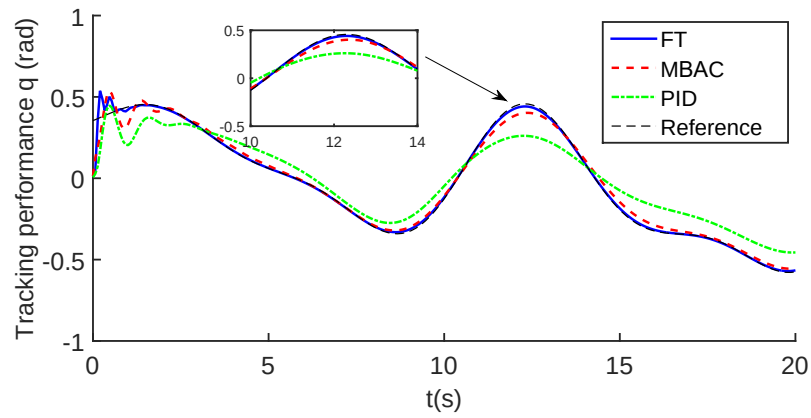


Figure 9: Tracking performance of Joint 3

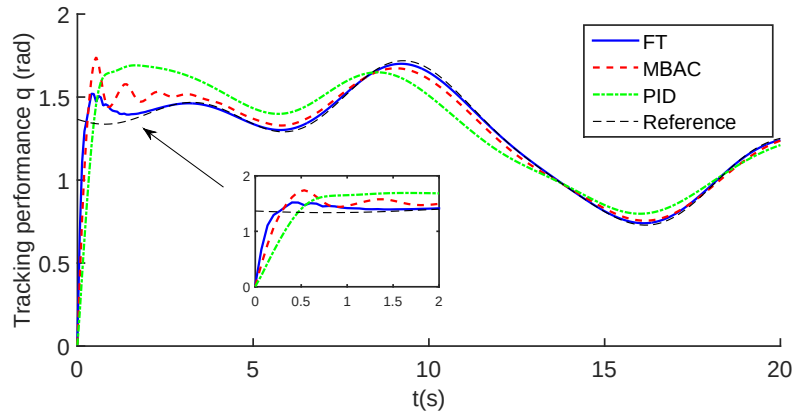


Figure 10: Tracking performance of Joint 4

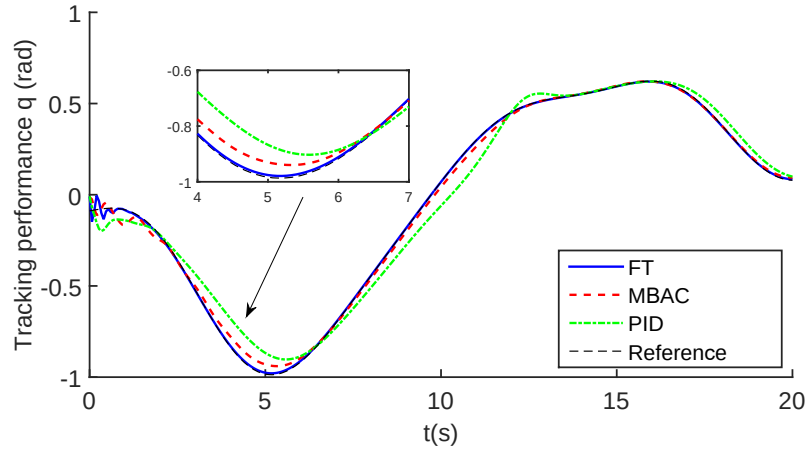


Figure 11: Tracking performance of Joint 5

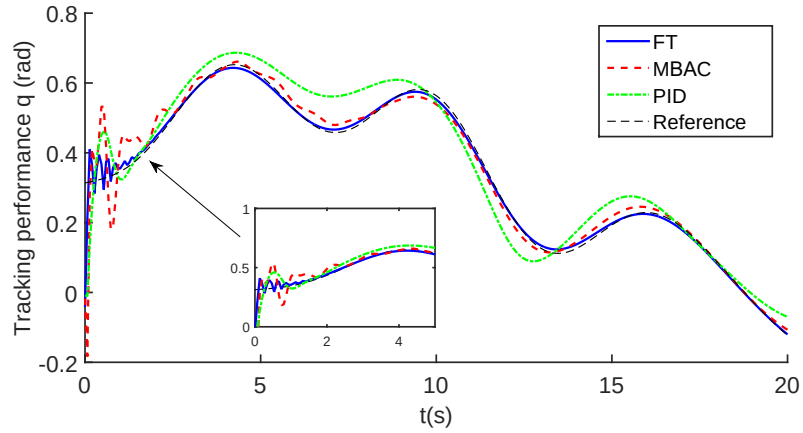


Figure 12: Tracking performance of Joint 6

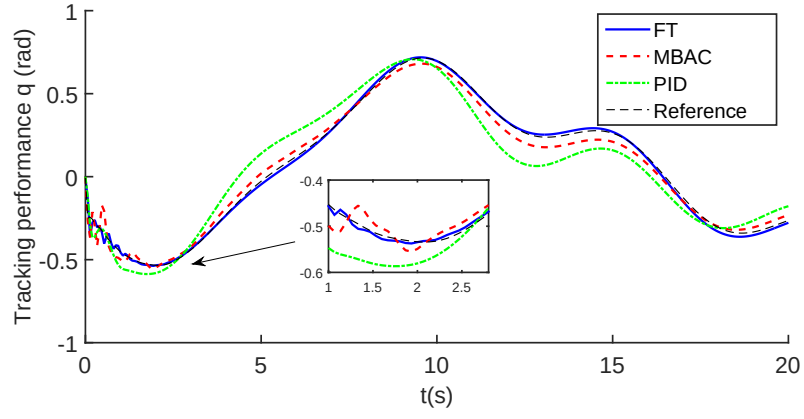


Figure 13: Tracking performance of Joint 7

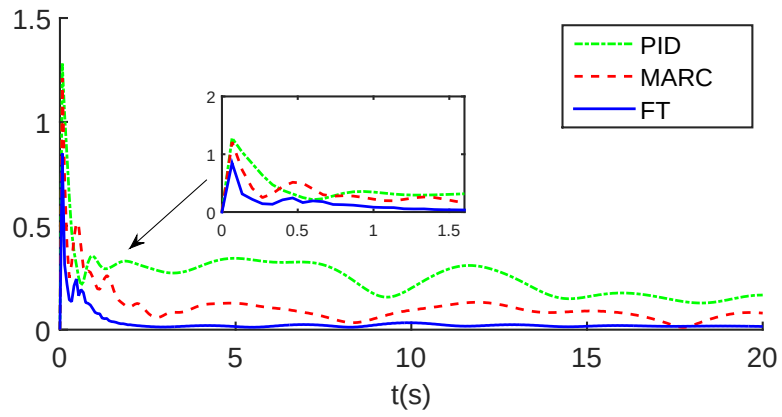


Figure 14: RMSE of the tracking error

Table 1: The Estimation Performance of Dynamic Parameters

No.	Real value	Proposed	Adaptive	No.	Real value	Proposed	Adaptive
1	0.0689	0.1322	0.0864	23	0.0208	0.0205	0.0084
2	-0.2773	-0.6104	-1.4653	24	0.0118	0.0110	0.0105
3	-4.1237	-4.0514	-3.4160	25	0.0021	0.0032	0.0338
4	1.5528	1.5552	1.3533	26	0.0284	0.0283	0.0063
5	0.6692	0.9955	1.8549	27	-0.0011	-0.0015	0.0344
6	1.5244	1.5274	1.4008	28	0.0033	0.0032	-0.0114
7	-0.0253	0.0211	0.1150	29	0.0093	0.0091	0.0345
8	0.1696	0.1666	0.0683	30	0.0071	0.0069	0.0000
9	0.4057	0.3986	0.4159	31	0.0007	0.0013	0.0047
10	0.0168	0.0600	0.0580	32	0.0168	0.0166	-0.0086
11	-0.0241	-0.0670	-0.1071	33	0.0006	-0.0007	0.0146
12	0.0087	0.0334	0.0360	34	0.0112	0.0110	0.0016
13	0.0027	-0.0051	0.0484	35	0.0054	0.0055	0.0021
14	0.0332	0.0839	0.0266	36	0.0039	0.0031	0.0226
15	-0.4721	-0.4638	1.4161	37	-0.0002	-0.0008	0.0109
16	-1.8565	-1.8240	-1.1679	38	0.0028	0.0027	0.0088
17	-0.0162	-0.0153	-0.0279	39	0.0005	0.0005	0.0013
18	0.7035	0.6918	0.4194	40	0.0009	0.0009	-0.0030
19	0.1869	0.1836	-0.2221	41	0.0005	0.0007	0.0044
20	0.6905	0.6790	0.3696	42	0.0001	0.0001	0.0179
21	-0.0246	0.0217	0.1070	43	0.0002	0.0002	0.0117
22	0.0087	0.0085	-0.0214				

proposed FT method. Fig. 14 show the profile of the root mean square error (RMSE) of tracking error of the 7 joint. We can see clearly from the figure that, both steady state and transient errors are improved than the other twos, which have show the effectiveness of the proposed controller.

Comparisons of the parameters estimation performance have also been carried out based on our proposed FT estimated algorithm and the adaptive estimation method in [27]. The detailed of the estimated parameters are shown in Table III, where the real value of the dynamic parameters, the parameters estimated by using our proposed finite-time algorithm and adaptive estimation method in [27] have been presented, respectively. We can see from the Table III that, the parameters estimated by using our proposed method is close to the real dynamic parameters, and has more estimation accuracy than conventional adaptive estimation method.

As suggested, the above discussions have been presented in Section IV, Page 8 of the revised manuscript.

Response to Referee No. 4:

Manuscript No.17-TIE-2621

**“Adaptive Parameter Estimation and Control Design for Robot Manipulators
with Finite-Time Convergence”**

The authors would like to express their sincere appreciation to the referee for his/her time and efforts to have our work reviewed and evaluated, and the recognition of the contribution of the paper.

1. *Comments: “Paper presents a new approach for the motion control and parameter estimation of a robot manipulator. ”*

Response: Thanks for your recognition of our work.

2. *Comments: “The following comments are in order: I detected multiple grammar errors and typos. Paper should experience proof reading before resubmitting.”*

Response: The authors sincerely thank the reviewer for the helpful comment. According to your suggestion, we have carefully proof-read the paper and tried our best to correct typos and grammar errors in the manuscript. In addition, we have also invited a native English-speaker to help us polishing this paper. For your convenience, the modifications are highlighted in blue color in the revised manuscript.

2. *Comments: “Figure 1 does not make clear what are the inputs and outputs for each subsystem. ”*

Response: Thanks for your constructive comments to improve our paper. As suggested by the reviewer, Figure 1 has been modified and updated in the revised manuscript. Please refer to Section II, Page 3 for more details.

3. *Comments: “ Simulation results are a clear disadvantage of the paper. Authors are encouraged to present real time experimental results in some two degrees of freedom mechanism with the aim to prove the practical viability of the method.”*

Response: Thank you very much for your constructive comments, which are appreciated. According to your suggestion, experiment results are performed based on a Baxter robot to further verify the effectiveness of the proposed kinematics estimation algorithm. For your convenience, the experiment results are depicted as below, and please refer Section IV, Page 7 for more details.

the update text of the experiment results

In addition, experiments have been further performed based on the left robotic arm of a real Baxter robot, as shown in Fig.3. In the experiment, the robot is commanded to track a set of sinusoidal and cosine trajectories depicted in Fig.14. The estimation results are shown in Fig.15. We can see from the figure that, most of the estimated parameters are close to

the nominal values (dashed line) with satisfactory convergence performance. In terms of the above test results, the robustness of the proposed estimation algorithm can be verified.

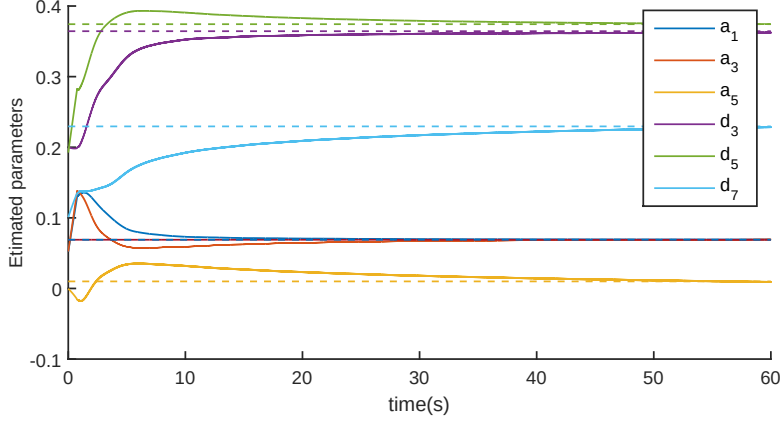


Figure 15: The kinematics estimation performance

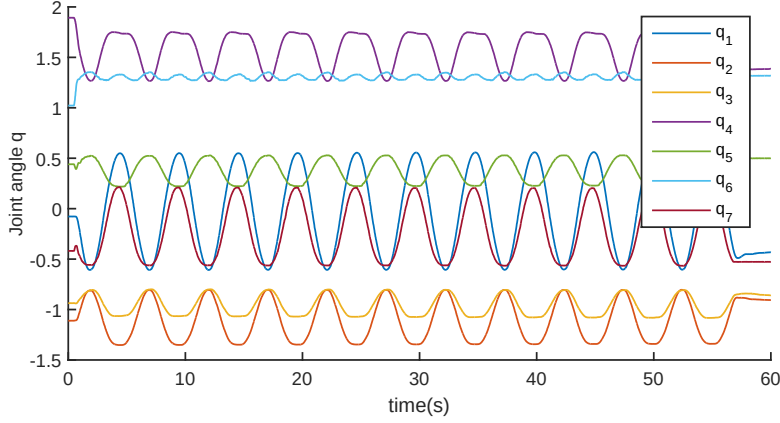


Figure 16: The trajectories of the joint angles

4. *Comments: "Authors are encouraged to perform comparisons taking into account the encoder quantification (or camera sensor), delay in the output measurements and dirty derivative velocity estimations. See, for example, – Moreno-Valenzuela, J., & Aguilar-Avelar, C. (2018). Motion Control of Underactuated Mechanical Systems, Springer-Verlag, Cham, Chapter: Identification of Underactuated Mechanical Systems. – Miranda-Colorado, R., & Moreno-Valenzuela, J. (2017). Experimental parameter identification of flexible joint robot manipulators. Robotica, 1-20."*

Response: The authors sincerely thank the reviewer for the constructive comments. According to your suggestion, the above mentioned state-of-the-art studies of robot identifications have been discussed and cited in the introduction part of the revised manuscript.

Please refer to Section I, Page 1 for more details. Additionally, to further show the effectiveness of our proposed estimation algorithm, robustness tests are conducted by adding unknown external disturbances to the robot system model. The test results show that the estimated parameters are observed with satisfactory convergence rate and accuracy, even in the presence of external disturbances. The details about robustness tests could be found in Section IV, Page 7 in the revised manuscript.

5. *Comments: “Explain why one should use the adaptive control law given in the paper. There a lot of adaptive controllers providing acceptable performance. ”*

Response: Thanks for your constructive comment, which is appreciated. In the past decades, adaptive control has been well investigated because of its online learning ability to update control parameters and to handle model uncertainties, e.g., in the model based control, it has been successfully employed to estimate the unknown parameters. In most exiting reference, however, the adaptive laws are designed to minimize the error of tracking control based on the gradient descent method. From this point of view, the parameter estimation convergence speed of gradient based adaptive laws could be regarded a function of the observer error and control error, which create strong couplings. This implies that the convergence of the estimated parameters may be effected by the system tracking errors. To improve the estimation performance, in this paper, we design a finite time adaptive estimation scheme with a leakage term driven by parameter estimation errors, such that precise and fast estimation of robotic parameters can be retained by using the information estimation error. Comparing with traditional gradient descent based adaptive method, the proposed adaptive estimation algorithm can guarantee the finite-time identification by using the information of the parameters estimated errors with better control performance from the fast convergence of the estimated parameters.

6. *Comments: “Write a table with the parameters estimated by different identification algorithms.”*

Response: Thanks for your constructive comments to improve our paper. According to your suggestion, comparative studies by using different estimation methods have been carried out and presented in the revised manuscript. The detailed of the estimated parameters are shown in Table II, where the real values of the dynamic parameters, the parameters estimated by using our proposed finite-time algorithm and the adaptive estimation method have been presented, respectively. For your convenience, the added text is listed as below, please refer to Section V, Page 8 for details.

the update text of the comparisons

Comparisons of the parameters estimation performance have also been carried out based on our proposed FT estimated algorithm and the adaptive estimation method in [27]. The detailed of the estimated parameters are shown in Table III, where the real value of the dynamic parameters, the parameters estimated by using our proposed finite-time algorithm and adaptive estimation method in [27] have been presented, respectively. We can see from

Table 2: The Estimation Performance of Dynamic Parameters

No.	Real value	Proposed	Adaptive	No.	Real value	Proposed	Adaptive
1	0.0689	0.1322	0.0864	23	0.0208	0.0205	0.0084
2	-0.2773	-0.6104	-1.4653	24	0.0118	0.0110	0.0105
3	-4.1237	-4.0514	-3.4160	25	0.0021	0.0032	0.0338
4	1.5528	1.5552	1.3533	26	0.0284	0.0283	0.0063
5	0.6692	0.9955	1.8549	27	-0.0011	-0.0015	0.0344
6	1.5244	1.5274	1.4008	28	0.0033	0.0032	-0.0114
7	-0.0253	0.0211	0.1150	29	0.0093	0.0091	0.0345
8	0.1696	0.1666	0.0683	30	0.0071	0.0069	0
9	0.4057	0.3986	0.4159	31	0.0007	0.0013	0.0047
10	0.0168	0.0600	0.0580	32	0.0168	0.0166	-0.0086
11	-0.0241	-0.0670	-0.1071	33	0.0006	-0.0007	0.0146
12	0.0087	0.0334	0.0360	34	0.0112	0.0110	0.0016
13	0.0027	-0.0051	0.0484	35	0.0054	0.0055	0.0021
14	0.0332	0.0839	0.0266	36	0.0039	0.0031	0.0226
15	-0.4721	-0.4638	1.4161	37	-0.0002	-0.0008	0.0109
16	-1.8565	-1.8240	-1.1679	38	0.0028	0.0027	0.0088
17	-0.0162	-0.0153	-0.0279	39	0.0005	0.0005	0.0013
18	0.7035	0.6918	0.4194	40	0.0009	0.0009	-0.0030
19	0.1869	0.1836	-0.2221	41	0.0005	0.0007	0.0044
20	0.6905	0.6790	0.3696	42	0.0001	0.0001	0.0179
21	-0.0246	0.0217	0.1070	43	0.0002	0.0002	0.0117
22	0.0087	0.0085	-0.0214				

the Table III that, the parameters estimated by using our proposed method are close to the real dynamic parameters, and has more estimation accuracy than conventional adaptive estimation method.